

SYED MAOZ ARIF

Hardware Developer & Electronics Engineer

Acting Vice President @ IET On Campus CUI Abbottabad

syedmaoza@gmail.com | +92 321 914 3144 | [LinkedIn](#) | [Portfolio](#) | Abbottabad, Pakistan

PROFESSIONAL SUMMARY

Hardware developer and electronics engineering student (4th semester) at COMSATS University with hands-on experience in multi-layer PCB design, embedded systems, and IoT. Builder of 9 real-world projects, holder of 17 certifications, and winner of 5 awards including a global Instructables contest. Active freelancer on Fiverr & Upwork serving international clients. Acting Vice President at IET On Campus and Cabinet Member of the Youth Parliament Pakistan.

EDUCATION

BSc Electrical and Electronics Engineering (EEE)

2024 – 2028 (Expected)

COMSATS University Islamabad, Abbottabad Campus

4th Semester

- Student ID: FA24-EEE-014 | Focus: Embedded Systems, PCB Design, Signal Integrity, IoT
- Altium Global Scholarship Awardee (March 2026)

FSc Pre-Engineering

Completed 2024

Peace Group of School & College, Abbottabad

Higher Secondary

- Core Subjects: Mathematics, Physics, Analytical Reasoning

LEADERSHIP & PROFESSIONAL EXPERIENCE

Acting Vice President

July 2026 – Present

IET On Campus, CUI Abbottabad

Leadership

- Guiding chapter operations and representing the student branch to foster engineering excellence
- Organizing technical events and professional development initiatives for 50+ members

Membership Head

May 2025 – July 2026

IET On Campus, Abbottabad

Leadership

- Led membership initiatives and organized technical workshops (Computer Vision & AI in UAVs)
- Commended for extraordinary leadership, team management, and problem-solving abilities

Cabinet Member

2025 – 2027 Session

Youth Parliament Pakistan

Governance

- Serving in a leadership capacity for youth advocacy and national policy discussions

Freelance Hardware & Mechanical Designer

Ongoing

Fiverr & Upwork

Professional

- Professional multi-layer PCB design, schematic capture, and 3D CAD modeling for international clients
- Project scoping, deadline management, and professional client communication

KEY PROJECTS

IncliSense Precision Digital Level

2026 | Runner-Up, Instructables Sensors Contest (Global)

Arduino Nano | MPU6050 | IoT | 3D Printing

- Precision leveling device with dual-axis spatial data, 11-LED interface, integrated buzzer, and PC telemetry
- Custom 3D-printed ABS enclosure; documented and published on Instructables

Industrial Edge AI Condition Monitoring & 3D Digital Twin

Edge AI | Condition Monitoring | 3D Digital Twin | IoT

- Complex Engineering Problem project; secured 1st Position in Software Category at COMSATS University

First FPGA Programming – iCEbreaker

2026

Verilog | iCEbreaker FPGA | Digital Logic | iceStorm Toolchain

- LED blinking via Verilog HDL; learned digital logic synthesis, pin assignment, and bitstream generation

6S UAV Battery Management System (BMS)

2026

BQ76930 | XT60 | MOSFETs | UAV Architecture

- Complete 6-cell BMS schematic with charge/discharge protection MOSFETs and hardware telemetry

Mixed-Signal RF Transceiver Board

2026

ESP32 | RF TX/RX | 50Ω Impedance Matching

- 50Ω impedance-matched RF traces, unified ground plane with 312 stitching vias (Faraday cage)

Smart Lighting System with Obstacle Detection	2026
Altium Designer ATMEGA328PB-AU VL53L0X ToF TP4056 USB-C	
• Custom PCB with Time-of-Flight sensor, dual LED arrays, and USB-C battery charging circuit	
Omni-Sense Node: Mixed-Signal DAQ	2026
ESP32 SPI PWM RFID MQ-Series Sensors	
• Centralized DAQ system with 15 heterogeneous peripherals; dual-rail power distribution network	
Wi-Fi Router UPS Power Bank	2026
Hardware Design BMS DC-DC Conversion 3S Li-ion	
• Custom DC UPS with 7-segment voltage display, rotary adjustment, and fused overcurrent protection	
Personal Portfolio Website	2026
HTML5 CSS3 JavaScript Firebase Firestore UI/UX	
• Responsive glassmorphism design with Firebase social features, AI assistant mode, and GPU-optimized animations	

AWARDS & HONORS

• Runner-Up – Instructables Sensors Contest 2026 (Global)	— Bronze Medal for “IncliSense” project	April 2026
• 1st Position – Best CEP Project Award	— COMSATS University & District Youth Office Abbottabad	July 2026
• 1st Position – Best Circuit Design	— COMSATS University & IET Hardware Workshop	Feb 2026
• Altium Global Scholarship Awardee	— Selected for emerging engineering excellence	March 2026
• Best Project Award	— COMSATS University – Outstanding engineering design & innovation	Feb 2026

CERTIFICATIONS (17)

• Altium Certified: Designer Essentials	• IET On Campus Recognition
• Altium Education PCB Basic Design	• IET Experience Certificate
• Generative AI Mastermind (Outskill)	• Student Week SP26 Certificate
• Arduino & IoT – Cardiff Institute	• Kangaroo Test Achievement
• Unleashing Creativity with AI	• Best Project Award Certificate
• Practical Entrepreneurship	• Inovatex Project 2025
• 3D Printing Certificate	• Best CEP Project Award – 1st Position
• Solar Solution for Pakistan	• Organizer – Project Exhibition 2026
• IET On Campus Leadership	

TECHNICAL SKILLS

PCB & Hardware	Altium Designer, Altium 365, CircuitStudio, KiCad, EasyEDA, Proteus, LTspice, Multisim
Embedded & IoT	ESP32, STM32, Arduino, ATMEGA328PB, Verilog (FPGA), MQTT, Firebase, Blynk
3D CAD	Autodesk Inventor, SolidWorks, Fusion 360, AutoCAD, Tinkercad
Programming	Python, MATLAB, C/C++ (Embedded), Verilog, HTML5, CSS3, JavaScript
Documentation	LaTeX, Overleaf, draw.io, Microsoft Office Suite
AI Tools	Gemini, ChatGPT, Claude, Google AI Studio, Microsoft Copilot
Design & Media	Adobe Photoshop, Canva, Inkscape

INTERPERSONAL SKILLS

Public Speaking		Leadership & Team Management		Communication & Client Relations		Collaboration & Empathy
Critical Thinking & Problem Solving		Project Scoping & Time Management				

INTERESTS

Cricket		Trekking		Exploring New Places
---------	--	----------	--	----------------------